Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2013-0451-DS-3-RF-101	Issued	2013-12-09

UT Thickness Data Sheet - 3-RF-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 3-RF-101 located within the ATU-100. The scope of work involves the re-measurement of the reflux line to ensure accurate wall thickness data following the correction of instrument velocity settings.

Kevin Doucette, acting as the lead UT technician and SNT-TC-1A UT Level II for Halward Inspection Services Ltd., is responsible for the execution of the measurements and the recording of the data presented herein.

Technical Basis and Standards

The inspection was performed in accordance with the requirements of ASTM E797. All thickness measurements were conducted using the stainless-steel velocity setting appropriate for A312 TP316 material.

The field activities were performed under the guidelines of HIS-UT-002 Rev 1 and QSCP-NDT-001 Rev 2. The latter procedure incorporates specific requirements for the CUI program, including insulation removal and the use of inspection ports.

Line Design Parameters

Line 3-RF-101 is specified as A312 TP316 with an outer diameter of 88.9 mm and a nominal wall thickness of 5.49 mm. The design pressure is 517 kPa(g) and the design temperature is 104 °C.

The minimum allowable wall thickness (t-min) for this line is 2.41 mm. Any measured thickness falling below this retirement threshold shall be documented as an NCR.

Inspection Results and Reporting

The data recorded in this sheet supports the findings detailed in HIS-2013-0451, which documents the re-measurement of reflux lines 3-RF-101 and 2-RF-102 using the correct velocity settings.

Thickness readings are used to determine the corrosion rate and remaining life. Based on the Class 2 designation for reflux lines, the maximum thickness inspection interval is 10 years and the external visual interval is 5 years.

Thickness readings, measured 2013-12-09

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-09-09	5.39	5.37	0.006	0.014	>50	2023-12-09	
3-RF-101-02	5.49	2.41	2013-09-09	5.34	5.40	0.003	-0.004	>50	2023-12-09	
3-RF-101-03	5.49	2.41	2013-09-09	5.45	5.37	0.005	0.010	>50	2023-12-09	
3-RF-101-04	5.49	2.41	2013-09-09	5.33	5.41	0.003	-0.010	>50	2023-12-09	
3-RF-101-05	5.49	2.41	2013-09-09	5.26	5.32	0.007	0.010	>50	2023-12-09	
3-RF-101-06	5.49	2.41	2013-09-09	5.25	5.17	0.020	0.030	>50	2023-12-09	
3-RF-101-07	5.49	2.41	2013-09-09	5.43	5.48	-0.003	-0.017	>50	2023-12-09	
3-RF-101-09	5.49	2.41	2013-09-09	5.12	5.26	-0.562	-0.562	>50	2023-12-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-09-09	3.71	3.60	0.018	0.032	44.9	2023-12-09	
2-RF-102-02	3.91	2.16	2013-09-09	3.65	3.70	0.010	0.023	>50	2023-12-09	
2-RF-102-03	3.91	2.16	2013-09-09	3.76	3.79	0.008	0.001	>50	2023-12-09	
2-RF-102-04	3.91	2.16	2013-09-09	3.80	3.82	0.003	-0.001	>50	2023-12-09	
2-RF-102-05	3.91	2.16	2013-09-09	3.69	3.67	0.080	0.080	18.8	2023-05-05	
2-RF-102-06	3.91	2.16	2013-09-09	3.71	3.67	0.161	0.161	9.4	2018-08-22	
2-RF-102-07	3.91	2.16	2013-09-09	3.82	3.72	0.014	0.017	>50	2023-12-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-12-09	_____